

Operation Ghost Ledger – Digital Forensics Capstone

Course: Advanced Digital Forensics / ISS 7133

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Case summary

Senator Kwame Nkosi (Chair, Senate Committee on Public Works) and Director-General Amara Okafor (National Water & Power Authority) are accused of soliciting a 15% kickback on a \$500M public infrastructure contract — the Bahari River Dam. Evidence leaked to a whistleblower site and went viral. The defence alleges the leak is an AI-driven political framing operation (voice cloning, AI-generated financial records, planted documents).

The question this investigation must answer: does the digital evidence support genuine corruption, digital manipulation, or a mixture of authentic and manipulated material?

Core hypothesis

The leaked files are a mixture of authentic and AI-manipulated content.

Sub-hypotheses

1. The audio base is authentic, but the 15% figure was spliced.
2. The PDF contains authentic structural data but doctored financial amounts.
3. The spreadsheet was created on Okafor's laptop and later modified.

Falsification criteria — the hypothesis is weakened/falsified if:

- no splicing is found in the audio,
- metadata does not differ between relevant file versions, and/or
- no AI-generation artifacts are detected in any exhibit.

See [01_case_setup/README.md](#) for the full case identification and evidence-package breakdown.

Repository structure

Folder	Contents
01_case_setup/	Case identification, core hypothesis, evidence inventory
02_phase1_public_validation/	Milestone 1 — Package A validation, hash catalogue

Folder	Contents
03_phase2_acquisition/	Milestone 2 — Package B imaging, chain of custody
04_phase3_analysis/	Weekly case files — filesystem, memory, network, mobile, cloud, anti-forensics
05_phase4_correlation/	Milestone 3 — master timeline, correlation report
06_phase5_final/	Milestone 4 — affidavit and capstone presentation
scripts/	Reproducible helper scripts (SHA-256 cataloguing)
METHODOLOGY.md	Daubert factors, cognitive-bias controls, tooling plan
Operation_Ghost_Ledger_Complete_Capstone.docx	Full narrative report across all five phases

Roadmap

Phase	Weeks	Deliverable
1 — Public Evidence Validation	1–3	Milestone 1 — Validation Report
2 — Hardware Acquisition & Imaging	4–5	Milestone 2 — Acquisition & Chain-of-Custody Log
3 — Multi-Domain Analysis	6–12	Weekly Case File Updates
4 — Correlation & Reconstruction	13–14	Milestone 3 — Correlation & Reconstruction Report
5 — Final Affidavit & Expert Testimony	15	Milestone 4 — Final Affidavit & Oral Presentation

Evidence handling

Do **not** upload confidential/raw forensic evidence — E01 images, RAM dumps, PCAPs, phone backups or cloud logs — to a public repository unless the instructor explicitly authorizes it.

Upload only approved derived artifacts: sanitized logs, reports, screenshots and scripts.

Academic integrity

Actual hashes, timestamps, IP addresses and forensic tool outputs must be generated from the real evidence package — they must **never** be invented. Where this repository shows **TBD** or [\[populate from analysis\]](#),

that field is intentionally left for the examiner to fill in from the actual supplied evidence.

Final submission checklist

- ☐ Milestone 1 — Public Evidence Validation Report
- ☐ Milestone 2 — Acquisition & Chain-of-Custody Log
- ☐ Weekly Case File Updates for Phase 3
- ☐ Milestone 3 — Correlation & Reconstruction Report
- ☐ Milestone 4 — Final Affidavit
- ☐ 15-minute oral presentation
- ☐ No fabricated hashes, timestamps, IPs, screenshots or conclusions
- ☐ All sensitive/original evidence excluded from this public repository